Formulas & Functions

formulas & operators

→ Used to perform calculations in Excel.

Arithmetic operators

① addition (+) : Used to add values

Ex :- =A1+B1

② Subtraction (-) : Used to subract values

Ex :- = A1 - B1

③ Multiplication (*): Used to multiply values

Ex : =A1*B1

④ Division (/) : Used to divide values

Ex : = A1/B1

⑤ Exponentiation (^) : used to raise power

Ex :- = A1^B1

⑥ Concatenation (&) : used to join text

Ex :- = A1&" "&B1

Note :- operator precedence

1. ^
2. * and /
3. + and -
4. &
5. Comparision operators.

Comparision Operators

1. = Ex:- = A1=B1

2. > Ex! = A1>B1

3. < Ex:- = A1<B1

4. >= Ex: = A1>=B1

5. <= Ex:- = A1<=B1

6. <> Ex! = A1<>B1 (not equal to)

Cell Referencing

→ used to refer to cells in formulas

Types:
- Relative reference
- Absolute reference
- Mixed reference

Relative reference - default

Ex:- A1

→ Changes when dragged A1 → A2 → A3

Absolute reference

→ locks row and column

Ex:- A1

→ doesnot change when dragged

Use: Tax rate, fixed values

Mixed reference

→ locks either row or column

lock column - Ex: $A1 (column fixed row changes)

lock row - Ex! A$1 (Row fixed column changes)

Note:- F4 shortcut (used to lock cell references)

A1 $\xrightarrow{\text{Press F4}}$ A1 $\xrightarrow{\text{Press F4}}$ A$1 $\xrightarrow{\text{Press F4}}$ $A1 $\xrightarrow{\text{Press F4}}$ A1 cycle repeats

functions intro

→ Predefined formulas

→ used to perform calculations quickly

① Average

Syntax : = AVERAGE (A1 : A10)

② Sum

Syntax : = SUM (A1 : A10) ignores blank cells & text values.

③ Minimum

Syntax : = MIN (A1 : A10)

④ Maximum

Syntax : = MAX (A1 : A10)

⑤ Count only numbers

Syntax : = COUNT (A1 : A10)

⑥ Count all non empty cells (numbers, text, dates etc..)

Syntax : = COUNTA(A1 : A10)

⑦ Count blank cells

Syntax : = COUNTBLANK (A1 : A10)

⑧ IF

Syntax : = IF (condition , value-if-true , value-if-false)

Ex :- = IF (A1 > 50 , "Pass", "Fail")

⑨ AND

Syntax : = AND (A1 > 50 , B1 > 50)

⑩ OR

Syntax : = OR (A1 > 50 , B1 > 50)

⑪ LEN

→ returns text length

Syntax: LEN(A1)

⑫ TRIM

→ removes extra spaces

Syntax: = TRIM(A1)

→ Use: data cleaning

⑬ CONCAT

→ combines text

Syntax: CONCAT (A1, B1)

IFS ()

Checks multiple conditions one by one and returns the value for the first TRUE condition

Syntax:

```
= IFS (
condition1 , result1,
Condition 2 , result2,
Condition3 , result3,
)
```

Ex: student grades

```
= IFS (
A2 > 90 , "A",
A2 > 80 , "B",
A2 > 70 , "C",
A2 > 60 , "D",
A2 < 60 , "Fail"
)
```

conditions checked from top to bottom

```
= IFS(
A2 > 70, "C",
A2 > 80, "B",
A2 > 90, "A"
)
```

wrong order

→ No inbuilt Else

So we can write like this :

```
= IFS (
  condition1 , result1 ,
  condition2 , result2 ,
  TRUE , default_result
  )
```

COUNTIF()

Counts the no.of cells that satisfy a specific condition.

Syntax : = COUNTIF (range , criteria)

Ex:- Count data analysts

= COUNTIF (A1 : A100 , "data Analyst")

Ex:- count blank cells

= COUNTIF (A : A , " ")

A:A means entire column in the excel.

Ex! count nonblank cells

= COUNTIF (A:A , "<>")

COUNTIFS ()

→ Counts cells / rows that satisfy multiple conditions

Syntax :-

```
= COUNTIFS ( criteria_range1 , criteria1 ,
             criteria_range2 , criteria2 ,
             ...)
```

Structure :- count rows where ;

Condition 1 is true AND
Condition 2 is true AND
ondition 3 is true

all conditions must be satisfied.

Ex :-

Role	City
data analyst	hyderabad
data analyst	bangalore
business analyst	hyderabad
data analyst	hyderabad

= COUNTIFS (A:A, "data analyst", B:B, "Hyderabad")

Result : 2

SUMIF ( )

→ adds value only when a specified condition is met.

Syntax :-

= SUMIF (range, criteria, sum-range)

range → where condition is checked

criteria → condition

sum-range → values to add

Ex :- Sum data analyst salaries

Role	Salary
data analyst	50000
business analyst	60000
Data analyst	70000

= SUMIF (A:A, "Data Analyst", B:B)

SUMIFS ( )

→ adds values that satisfy multiple conditions

Syntax :-

= SUMIFS (sum-range,
criteria-range 1, criteria1,
criteria-range 2, criteria2,
...)

Structure :- add values where

Condition 1 is TRUE AND
Condition 2 is TRUE AND
Condition 3 is TRUE

} all conditions must be satisfied

Ex:- Role | City | Salary

Role	City	Salary
data analyst	hyderabad	50000
data analyst	bangalore	60000
data analyst	hyderabad	70000
business analyst	hyderabad	8000

= SUMIFS (C:C, A:A, "data analyst",
B:B, "Hyderabad")

AVERAGEIF()

-> calculates the average of values that meet one condition.

Syntax:- =AVERAGEIF (range, criteria, average-range)

range -> where condition is checked

criteria -> condition

average-range -> range where average is to be done

AVERAGEIFS ()

-> calculates the average of values that meet multiple conditions

Syntax:-

= AVERAGEIFS (average-range,
criteria-range 1, criteria 1,
criteria-range 2, criteria 2)

LEFT, RIGHT, MID

-> These are called Text functions

-> They help extract specific characters from text

① LEFT() : returns characters from the left side of a string.

Syntax:- = LEFT (text, num-chars)

text -> cell containing text

num-chars -> no. of characters to extract

Ex:- A2 : Data Analyst

formula : LEFT (A2, 4)

Result : Data

② RIGHT( ) : returns characters from the right side of a text string.

Syntax : = RIGHT (text, num_chars)

Ex:- Extract last four digits

A1 : 11235461235011143

formula : = RIGHT (A1, 4)

Result : 1143

③ MID ( ) : Returns characters from the middle of a text string

Syntax:- = MID (text, start_num, num_chars)

text → Cell containing text

start_num → Position to start

num_chars → No. of characters

Ex:- A2 : EMP12345

formula : = MID (A2, 4, 5)

Result : 12345

UPPER, LOWER, PROPER (Text formatting functions)

Syntax	Meaning	
= UPPER(text)	all capital letters	ex: UPPER (A2)
= LOWER (text)	all small letters	ex: LOWER (A2)
= PROPER (text)	first letter capital	ex: PROPER (A2)

MINIFS()

-> returns smallest value that meets one or more conditions

Syntax:-

= MINIFS (min_range,
criteria_range1, criteria1,
criteria_range2, criteria2)

Example :-

Role	Salary
Data analyst	50000
data analyst	70000
Business analyst	60000

= MINIFS (B:B, A:A, "Data Analyst")

Result : 50000

Example :-

Role	City	Salary
data analyst	hyderabad	50000
data analyst	hyderabad	70000
data analyst	bangalore	45000

= MINIFS (C:C, A:A, "Data Analyst", B:B, "Hyderabad")

Result : 50000

MAXIFS()

-> returns the largest value that meets one or more conditions

syntax:-

= MAXIFS (max_range,
criteria_range1, criteria1,
criteria_range2, criteria2)

Note:- firstly excel only had AVERAGEIF(), COUNTIF(), SUMIF() and No MINIF() or MAXIF(), they used array formulas but later AVERAGEIFS(), COUNTIFS(), SUMIFS(), MINIFS() and MAXIFS() are invented.

MEDIAN ( )

→ returns middle value after sorting data

ex:- =MEDIAN(A1:A10)

→ less affected by outliers

STDEV (standard deviation)

→ measures spread of data

→ LOW STDEV : values close together

High STDEV : values spread apart

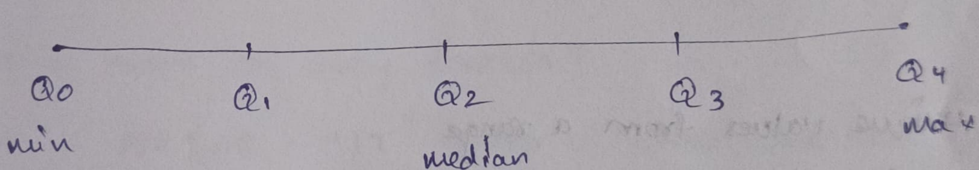
Types

STDEV.S (used for sample data)

STDEV.P (used for entire population)

QUARTILES

→ divide sorted data into four equal parts



Syntax :- =QUARTILE.INC ( )

=QUARTILE.EXC ( ) → Excludes minimum & Maximum values

MODE

→ returns most frequently occuring value

→ 2 types ← MODE.SNGL

MODE.MULT (if multiple modes present)

RANK

Syntax:- RANK (number, ref, [order]) → ascending / descending

number ↓ for which we have to give rank

ref ↓ with which reference

Array formulas

→ work with multiple values at once instead of a single value.
→ used for : filtering data, sorting data, returning multiple results

Dynamic array vs classic array

dynamic array is modern excel method. [Excel decides output range]

Steps :- 1. Select one cell
2. Enter formula
3. Press Enter

} Excel automatically returns results to multiple cells.

→ A blue /shawdow border appears around the output range
→ This is called spill range

classic array is an older excel method [you decide output range]

Steps :- 1. Select entire output range first
2. Enter formula
3. press Ctrl + shift + Enter

} Excel places formula in selected cells.

→ formula appears with { } around it

UNIQUE

→ used to return unique values from a range
→ removes duplicates automatically

syntax :- = UNIQUE (array)

SORT

→ used to sort data automatically
→ can sort in ascending or descending order.

syntax :- = SORT(array)

= SORT (array, [sort-index], [sort-order], [by-col])

array (required argument) → data to sort

sort-index (optional) → which column to sort by

sord-order (optional) → 1 for ascending (default) & -1 for descending.

by-col (optional) → FALSE for sort rows (default) & TRUE fort sort columns

MEDIAN with IF

→ used to calculate the median only for values that meet a condition.

Ex:

department	Salary
IT	50000
HR	40000
IT	60000
Sales	45000
IT	70000

To find the median salary of only IT employees:

= MEDIAN (IF (A2 : A6 = "IT", B2 : B6))

How it works:-

step 1 : IF checks condition

result :- 50000
FALSE
60000
FALSE
70000

Step 2 : Median only uses numeric values

Median : 60000

why is it called an array formula?

Because : A2 : A6 = "IT" doesn't check one cell. It checks multiple cells at once

MEDIAN + IF (Array formula) used to calculate median based on a condition.

Syntax : = MEDIAN (IF (condition - range = condition, value - range))

COUNT with SUMPRODUCT

→ Used to count records matching a condition

→ Before COUNTIFS existed, SUMPRODUCT was commonly used for conditional Counting

Ex:-

Month	Jobcount
Jan	10
Feb	15
Jan	20
Mar	12
Jan	18

To count how many records belong to January:

= SUMPRODUCT (- - (A2: A6 = "Jan"))

how it works

1. check condition

A2 : A6 = "Jan"

Result : TRUE
FALSE
TRUE
FALSE
TRUE

2. Convert TRUE/FALSE to 1/0

-- (A2: A6 = "Jan")

Result : 1
0
1
0
1

3. SUMPRODUCT adds values

1 + 0 + 1 + 0 + 1
= 3.

FILTER

Used to extract data based on conditions

Ex:-

Name	Sales
John	5000
Alice	8000
Bob	3000

Syntax:- = FILTER (array, include)

formula : (A2:B4, B2:B4 > 4000)

result :

John	5000
Alice	8000

SORTBY( )

→ Used to sort data based on another column/row.

Syntax : SORTBY (array, by_array)

Note :-

→ Use SORT when the column you want to sort by is inside the data you're returning. (sort this table by one of its own columns)

→ Use SORTBY when the column you want to sort by is outside the returned data or you don't want to display that column
[sort this data using another range]

lookup functions

→ used to search for a value and return related information from a table.

ex: Employee ID → Employee Name

Product ID → Price

use: search data, retrieve information.

VLOOKUP (vertical lookup)

→ used to search for a value in the first column & return a value from another column in the same row.

Syntax:- = VLOOKUP (lookup-value, table-array, col-index-num, [range-lookup])

look-up-value → value to search

table-array → Table containing the data (containing lookup column & return column(s))

col-index-num → column number to return (Ex: 1, 2, 3 ..)

range-lookup → choose FALSE for exact match & TRUE for approximate match

Limitations

1. Searches only left to right
2. lookup column must be first column
3. Uses hardcoded column numbers
4. less flexible than XLOOKUP

XLOOKUP

XLOOKUP is the modern replacement for VLOOKUP

→ used to search for a value & return related information from any column.

Syntax:- = XLOOKUP (lookup-value, lookup-array, return-array, [if-not-found])

lookup-value → value to search

lookup-array → column where Excel search.

return-array → column from which Excel returns result.

if-not-found → value to show if lookup value is not found

advantages

1. Can search left to right & right to left
2. No need to count column for col-index, can directly search & return.

HLOOKUP

H = Horizontal Lookup

→ Used to search for a value in the first row of a table and retu[rn] a value from another row in the same column.

Syntax :- = HLOOKUP (lookup-value, table-array, row-index-num, [range-loo[kup]]

arguments :-

lookup-value → value to search
table-array → table containing data
row-index-num = row number to return
range-lookup → FALSE for exact match & TRUE for approximate match

Ex :-

ID	101	102	103
Name	John	Alice	Bob
Salary	50000	60000	55000

formula : = HLOOKUP (102, A1:D3, 2, FALSE) → return second row. Result : Alice

Bucketing using XLOOKUP

→ Bucketing means grouping continuous values into categories.

Example: Salary

35000
65000
120000

Bucketing →

(A) Salary	(B) Category
35000	low
65000	medium
120000	high

first create a lookup table.

(C) Category	(D) min salary
low	0
medium	50000
high	100000

→ Create a column (B) for the categories & write formula like

= XLOOKUP (A2, D2:D4, C2:C4, , 1) → match mode
0 – exact
-1 – exact or <
1 – exact or >

MATCH function

→ used to find the position of a value in a range.

Syntax:- =MATCH (lookup-value, lookup-array, [match-type])

Ex:- Job title

Data Analyst

Data Engineer

Data Scientist

formula : = MATCH ("Data Engineer", A2 : A4 ,0)

Result : 2

Use: · find row position

INDEX function

→ used to return a value from specific position in a range.

Syntax:- INDEX (array, row_num)

Ex:- Job title

Data Analyst

Data Engineer

Data Scientist

formula: INDEX (A2 : A4, 2)

Result: Data Engineer

Use: Return value by position.

INDEX + MATCH

→ Used together to perform lookups

→ Before XLOOKUP, this was a popular alternative to VLOOKUP

→ Today XLOOKUP is simpler & more commonly used.

Text functions

TEXTJOIN()

→ Used to combine text from multiple cells with delimiter

Syntax :- = TEXTJOIN (delimiter, ignore_empty, text1, text2 ...)

delimiter → separator Excel places b/w values Ex , - |

ignore_empty → TRUE to ignore empty cells else false

text 1, text2 → values / cells to combine

FIND()

→ Used to find position of text inside another text.

Syntax : = FIND (find_text, within_text)

SEARCH ()

→ similar to find but not case sensitive

Syntax :- = SEARCH (find_text, within_text)

SUBSTITUTE ()

→ used to replace specific text

Syntax :- = SUBSTITUTE (text, old_text, new_text)

VALUE()

→ Converts text that looks like a number into actual numeric value

Syntax : = VALUE (text)

EXACT()

→ Compares two text strings and checks whether they are exactly the same.

Syntax :- = EXACT (text 1, text 2)

Delimiter :- character used to separate pieces of text

Ex :- ,
;
-
/

Date & Time functions

DAY()

→ used to extract day from date

Syntax: =DAY(date)

MONTH()

Syntax: =MONTH(date)

YEAR()

Syntax: =YEAR(date)

DATE()

→ used to create valid data

Syntax: =DATE(year, month, day)

TODAY()

→ returns current date.

Syntax: =TODAY()

HOUR()

→ Extracts hour from time

Syntax: =HOUR(time)

MINUTE()

=MINUTE(time)

SECOND()

=SECOND(time)

TIME()

→ creates valid time value

=TIME(hour, miulte, second)

TEXT()

→ used to display date/time in custom format

Syntax: =TEXT(value, "format")

Common formats:- DD-MM-YYYY

HH:MM

HH:MM:SS

DATEIF()

→ used to calculate difference between 2 dates

Syntax:- =DATEIF(start_date, end_date, u

Units

D - Days

M - Months

Y - Years

Use: Employee experience

Age calculation